1) $E[X] = 2 \wedge SD[X] = 1{,}5$

X = endring i gjennomsnittlig temp. (menneskeskapt)

Y = ——— " ——— (naturlig)

$E[Y] = 0 \quad SD[Y] = 1$

$Z = X + Y$

a) $E[Z] = E[X+Y] = E[X] + E[Y] = \underline{2} + 0$

b) $Var[Z] = Var[X] + Var[Y] + 2cov[X,Y]$

$= 1{,}5^2 + 1^2 + 0$

$= \underline{3{,}250}$

c) $g(z) = 150z^2 - 300z$

$E[g] = E[150z^2] - E[300z]$

$= 150E[z^2] - 300E[z]$

$$= 150\,E[Z^2] - 300 \cdot 2$$

$$Var[Z] = 3{,}25 = E[Z^2] - (E[Z])^2$$

$$E[Z^2] = 3{,}25 + 4 = 7{,}25$$

$$E[g(Z)] = 150 \cdot 7{,}25 - 300 \cdot 2$$

$$= \underline{\underline{487{,}2}}$$

2) Øyefarge A v a $\quad P(A) = 0{,}1$

Håfarge B v b $\quad P(B) = 0{,}8$

$P(A \cap B) = 0{,}08$

$$X \begin{cases} 0 & \text{for a} \\ 1 & \text{for } A \end{cases}$$

$$Y \begin{cases} 0 & \text{for b} \\ 1 & \text{for } B \end{cases}$$

a) $E[X] = \sum_x x f(x) = 0 \cdot 0{,}9 + 0{,}1 \cdot 1 = 0{,}100$

$P(A) = 0{,}1 \Rightarrow P(\bar{a}) = 0{,}9$

b) $Var[X] = E\left[\left(X - \mu_{g(x)}\right)^2\right]$

$= (0 - 0{,}1)^2 \cdot 0{,}9 - (1 - 0{,}1)^2 1$

$= 0{,}09 - 0{,}9^2 = 0{,}9 - 0{,}81 = 0{,}090$

c) $E[X \cdot Y] = E[X] \cdot E[Y]$

$E[Y] = 0 \cdot 0{,}2 + 0{,}8 \cdot 1 = 0{,}8$

$E[XY] = 0{,}8 \cdot 0{,}1 = 0{,}08$

d) $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

$\mu_X = E[X]$

$= 0$ (gener er uavhengige)

e) $= 0$

$Corr(XY) = \frac{Cov(XY)}{\sqrt{\quad}}$

3) $f_{XY} = \frac{2(2y+x)}{3}$, $0 \le x, y \le 1$

$$E[X] = \sum_x x \cdot f_x(x) = \int_0^1 x f_x(x)\,dx = \int_0^1 x \int_0^1 f_{XY}(x,y)\,dy\,dx$$

$$\int_0^1 f_{XY}(x,y)\,dy = \frac{2}{3}\left[y^2 + yx\right]_{y=0}^{1} = \frac{2}{3}(1+x)$$

$$\int_0^1 x \cdot \frac{2}{3}(1+x)\,dx = \frac{2}{3}\int_0^1 x^2 + x\,dx = \frac{2}{3}\left[\frac{1}{3}x^3 + \frac{1}{2}x^2\right]_{x=0}^{1}$$

$$= \frac{2}{3}\left(\frac{1}{3}+\frac{1}{2}\right) = \frac{2}{9} + \frac{2}{6} = \frac{5}{9}$$

$$Var[X] = E[X^2] - (5/9)^2$$

$$E[X^2] = \int_0^1 x^2 \frac{2}{3}(1+x)\,dx = \frac{2}{3}\int_0^1 x^2 + x^3\,dx$$

$$= \frac{2}{3}\left[\frac{1}{3}x^3 + \frac{1}{4}x^4\right]_{x=0}^{1} = \frac{2}{3}\left(\frac{1}{3}+\frac{1}{4}\right) = \frac{2}{3}\cdot\frac{7}{12} = \frac{7}{3\cdot 6} = \frac{7}{18}$$

$$Var[X] = \frac{7}{18} - \frac{25}{81} = \frac{7.45 - 25}{81} = \frac{31,5 - 25}{81} = \frac{6,5}{81}$$

$$= \frac{13}{162}$$

$$E(\bar{Y}) = \int_0^1 y \int_0^1 f_{XY}(x,y)\,dx\,dy = \int_0^1 y \cdot \frac{2}{3}(2y+\tfrac{1}{2})\,dy$$

$$= \frac{2}{3}\int_0^1 2y^2 + \frac{1}{2}y\,dy = \frac{2}{3}\left[\frac{2}{3}y^3 + \frac{1}{4}y^2\right]_{y=0}^{1} = \frac{2}{3}\left(\frac{2}{3}+\frac{1}{4}\right)$$

$$= \frac{4}{9} + \frac{1}{6} = \frac{5.5}{9} = \frac{11}{18}$$

$$Var[\bar{Y}] = \frac{2}{3}\int_0^1 y^2\left(2y+\tfrac{1}{2}\right)dy - (11/18)^2$$

$$= \frac{2}{3}\int_0^1 2y^3 + \frac{1}{2}y^2\,dy - 11/18^2 = \frac{2}{3}\left[\frac{1}{2}y^4 + \frac{1}{6}y^3\right]_0^1 - \frac{11}{18}^2$$

$$= \frac{2}{3}\left(\frac{1}{2}+\frac{1}{6}\right) - \frac{11}{18}^2 = \frac{1}{3} + \frac{1}{9} - \frac{121}{324} = \frac{108+36-121}{324} = \frac{23}{324}$$

$$Cov(X,Y) = E[X,Y] - E[X][E[X]$$

$$= \iint xy\,f_{xy}\,dy\,dx - \frac{5}{9}\cdot\frac{11}{18} \qquad \left| \frac{55}{162}\right.$$

$$= \frac{2}{3}\int xy(2y+x)\,dy$$

$$= \frac{2}{3}x\int_0^1 2y^2 + yx\,dy$$

$$= \frac{2}{3}x\left[\frac{2}{3}y^3 + \frac{1}{2}y^2x\right]_0^1$$

$$\rightarrow \int_0^1 \frac{2}{3}x\left(\frac{2}{3}+\frac{1}{2}x\right)dx$$

$$\int_0^1 \frac{4}{9}x + \frac{2}{6}x^2\,dx$$

$$\frac{4}{9}\int_0^1 x\,dx + \frac{2}{6}\int_0^1 x^2\,dx$$

$$\frac{4}{9}\left[\frac{1}{2}x^2\right]_{x=0}^{1} + \frac{2}{6}\left[\frac{1}{3}x^3\right]_{0}^{1} \Rightarrow \frac{4}{18} + \frac{2}{18} = \frac{6}{18}$$

$$\frac{6}{18} - \frac{55}{162} = \frac{6\times 18 - 110}{324} = \frac{108-110}{324}$$

$$-\frac{2}{324} = -\frac{1}{162}$$

g) $Var[Y-X] = Var[Y+(-X)] = Var[Y] + (-1)^2 Var[X]$

$$= \frac{13}{162} + \frac{23}{324} = \frac{49}{324}$$

h) $$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var[X]\cdot Var[Y]}} = \frac{-\frac{1}{162}}{\sqrt{\frac{13}{162}\cdot\frac{23}{324}}}$$

$$= -0,0817$$

4) MDG: 20%

Spår 13 pers. X = # som stämmer MDG

$$E[X] = 0{,}2 \cdot 13 = \underline{2{,}6}$$

5) $$\mathrm{Var}[X] = np(1-p)$$

$$= 13 \cdot 0{,}2(1-0{,}2) = \underline{2{,}08}$$

c) $P(\text{minst 1 MDG}) = P(X \geq 1)$

$$= 1 - P(X=0) = 1 - P(\text{Alla 13} \neq \text{MDG})$$

$$= 1 - \binom{13}{1} 0{,}8 \cdot (1-0{,}8)^{12}, \quad P(\text{MDG}') = 0{,}8$$

$$= \underline{0{,}945}$$

d) $$P(X=1 \mid X \geq 1) = \frac{P(X \geq 1 \mid X=1)\,P(X=1)}{P(X \geq 1)} = \frac{P(X=1)}{P(X \geq 1)}$$

$$= \frac{\binom{13}{1} 0{,}2 \cdot 0{,}8^{12}}{0{,}945} = 0{,}1890$$

5) 700 stud.

150 (P1) | 250 (P2) | 300 (P0)

$P(P1\text{-eler svarer på undersøkelse}) = 0,7$

$P(P2 \text{ — " — }) = 0,9$

$P(P0) = 0,6$

a) X = # svar på undersøkelse i P1

$E[X] = np = 150 \cdot 0,7 = 105$

b) Y = # svar i P1 + P2 + P0

$= np_1 + np_2 + np_0 = 150 \cdot 0,7 + 250 \cdot 0,9 + 300 \cdot 0,6$

$= 510,00$

6) $n=11$, $p=0{,}1$

a) $P(X=3) = \binom{11}{3} 0{,}1^3 (0{,}9)^{11-3} = 0{,}071027$

b) $P(X \le 3) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$

$= 0{,}9^{11} + 11 \cdot 0{,}1 \cdot 0{,}9^{10} + \frac{11 \cdot 10}{2} \cdot 0{,}1^2 \cdot 0{,}9^9 + 0{,}071027$

$= 0{,}98147$

c) $P(X \le 3 \mid Y \le 3) = P(X \le 3)$ | uavhengige

d) $P(X + Y = 2) = P(x=0 \cap y=2) + P(x=1 \cap y=1) + P(y=0 \cap x=2)$

$= 2P(x=0 \cap y=2) + P(x=1 \cap y=1)$

$= 2P(x=0) \cdot P(x=2) + P(x=1)^2$

$X \leftrightarrow Y$ (samme fordeling)

$= 2 \cdot 0{,}9^{11} \cdot \frac{110}{2} 0{,}1^2 0{,}9^9 + (11 \cdot 0{,}1 \cdot 0{,}9^{10})^2 = 0{,}28084$